

Video storyboard

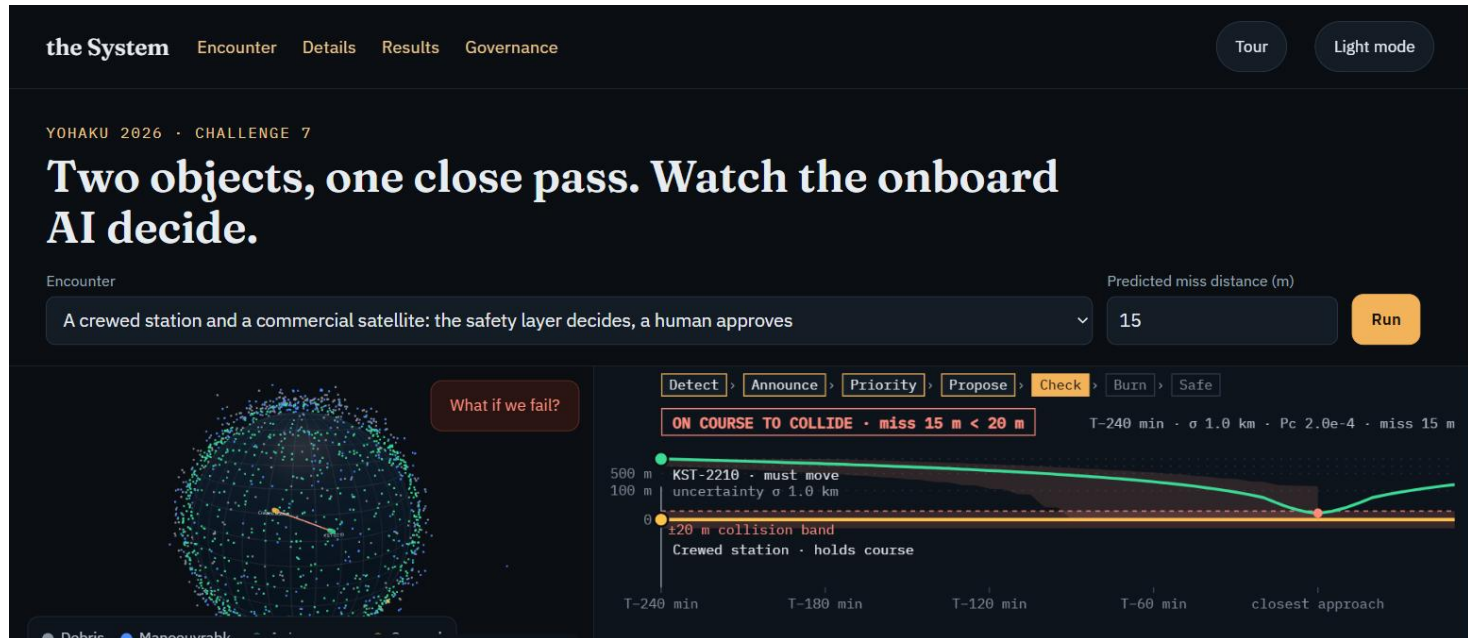
Eleven beats, 1:58 in total. Each page shows the screen to have on camera, what to click, what to point at, and the line to say. Record the screen at 1440×900 or larger in dark mode, playback speed Normal; load each page once before recording so the globe and fonts are cached; press Escape if the tour opens.

0:00 – 0:10	Encounter page · top
0:10 – 0:22	Plain title card (make it in your slides or a full-screen text)
0:22 – 0:30	Plain title card, second line
0:30 – 0:50	Encounter page · setup strip + stage (approach view)
0:50 – 0:58	Encounter page · the decision dialog
0:58 – 1:05	Encounter page · approach view after the burn
1:05 – 1:20	Encounter page · decision flow (scroll down)
1:20 – 1:28	Encounter page · “Or pick two real objects” (team server address)
1:28 – 1:35	Encounter page · approach view for the real pair
1:35 – 1:48	Details page · Results (nav: Details › Results)
1:48 – 1:58	Details page · Governance (nav: Governance)

If you run long, cut in this order: the last sentence of beat 3 (“So we added...”), then the ledger sentence in beat 9.

Beat 1 · 0:00 – 0:10

Encounter page · top



Do

- Open the Encounter page. Press Escape if the tour opens.
- Hold on the title for two seconds, then cut to the numbers card (next page) or keep the page and say the line.

Point at

- The title: Two objects, one close pass. Watch the onboard AI decide.
- Nothing to click yet.

Sav

“Two hundred thousand collision-avoidance manoeuvres in six months. In 2019, two satellites negotiated a near-miss by email, and one side never saw the last message.”

Beat 2 · 0:10 – 0:22

Plain title card (make it in your slides or a full-screen text)

What can we learn from mycelium?

sense locally · stay connected · reward partners that give back

Do

- Show the card full screen. No site on screen.

Point at

- Say the three words as they appear: sense locally, stay connected, reward partners.

Say

“The challenge asked what mycelium can teach us. Fungi coordinate with no boss: every tip senses locally, stays connected, and partners that give back are rewarded. We borrowed that architecture.”

Beat 3 · 0:22 – 0:30

Plain title card, second line

Not its morality.

priority · accountability · a human veto

Do

- Same card; reveal the second line.

Point at

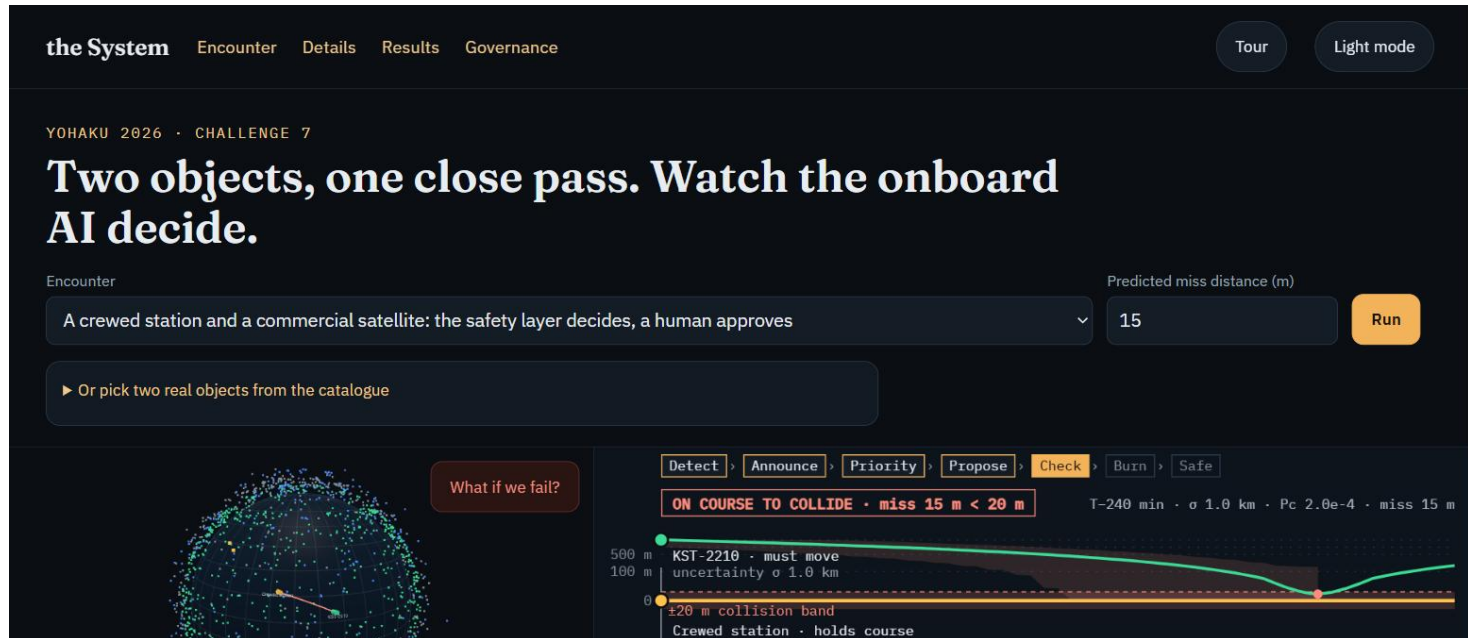
- The three additions: priority, accountability, a human veto.

Sav

“But not its morality. Fungi can't be held responsible; orbit can't repair itself. So we added priority, accountability and a human veto.”

Beat 4 · 0:30 – 0:50

Encounter page · setup strip + stage (approach view)



Do

- In Encounter, pick “A crewed station and a commercial satellite”, press Run, then Play.
- The step strip lights Detect > Announce > Priority; the dialog will open at the first step (next beat).

Point at

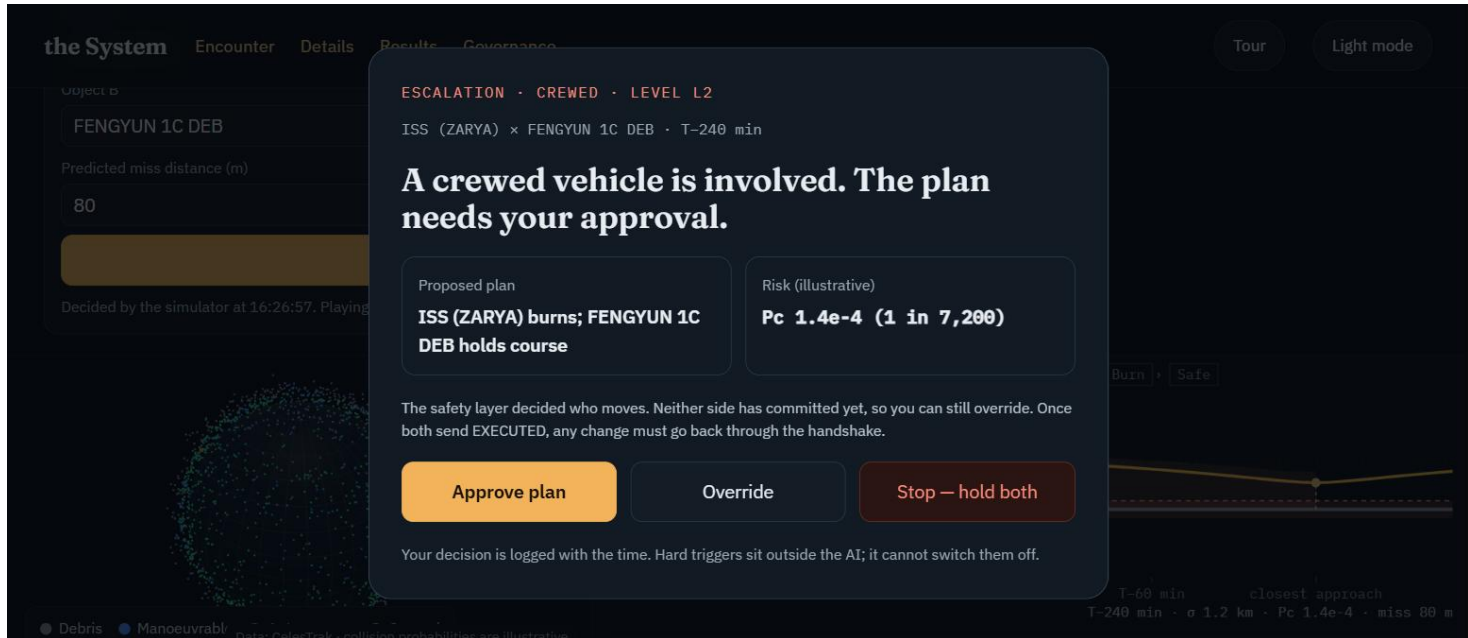
- The red verdict box: ON COURSE TO COLLIDE · miss 15 m < 20 m.
- The two paths: the yellow line holds course; the green path comes in and crosses inside the red ± 20 m band.
- The shaded funnel around the green path: tracking uncertainty, 1 km at first.

Say

“Every satellite carries the same small onboard AI. Watch one encounter: a crewed station and a commercial satellite, fifteen metres apart — on course to collide. Both announce themselves, and the safety layer decides from shared data: the crewed vehicle holds course, the other must move.”

Beat 5 · 0:50 – 0:58

Encounter page · the decision dialog



Do

- The dialog opens by itself. Hover the three buttons, then click Approve plan.

Point at

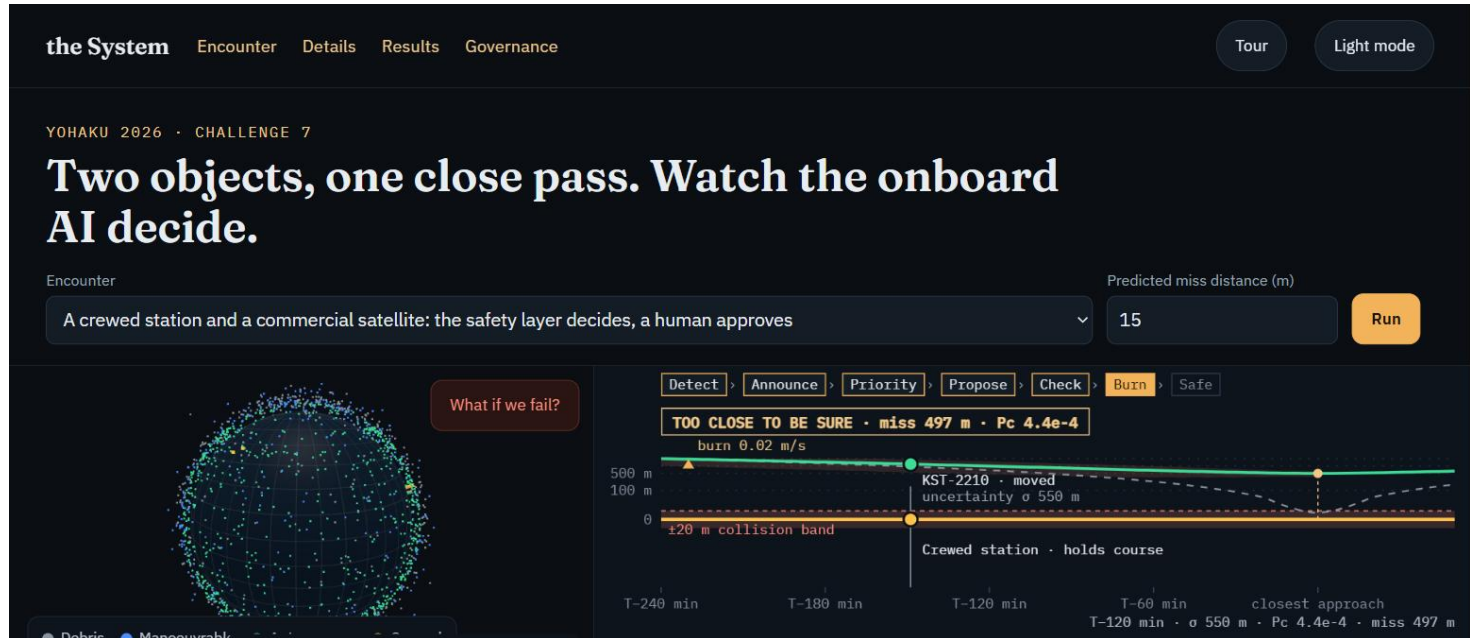
- ESCALATION · CREWED · LEVEL L2 and the proposed plan (who burns, who holds).
- Risk (illustrative) and the note: hard triggers sit outside the AI.

Say

“A crewed vehicle is involved, so the system stops and asks a human; the AI cannot switch that off. I approve.”

Beat 6 · 0:58 – 1:05

Encounter page · approach view after the burn



Do

- Let it play. Two steps later the burn happens (T-230).

Point at

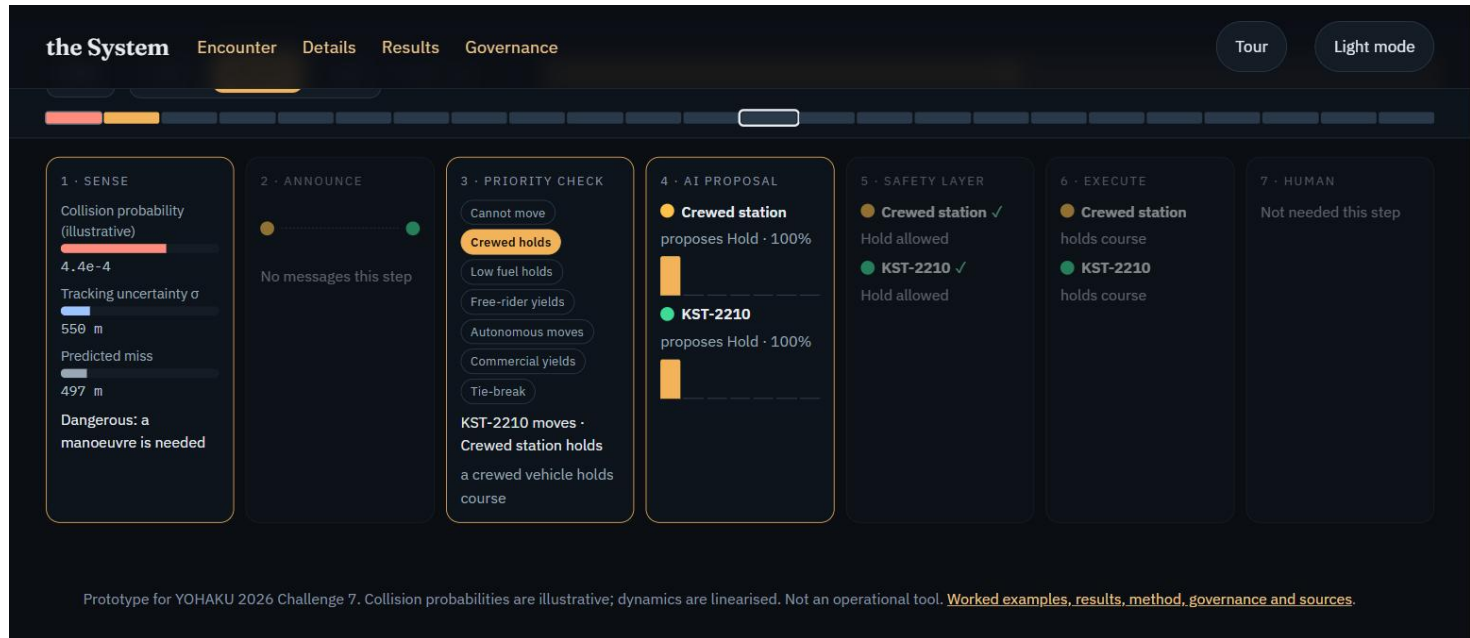
- The orange “burn 0.02 m/s” arrow.
- The old path turns into a faded dashed line into the band; the new green path stays at about 500 m.
- The verdict box changes: TOO CLOSE TO BE SURE while uncertainty is large, then SAFE.

Say

“The satellite burns early and small: its old path fades, the new one clears the collision band, and the verdict turns to safe.”

Beat 7 · 1:05 – 1:20

Encounter page · decision flow (scroll down)



Do

- Scroll down to the seven nodes. Move the mouse slowly left to right while they are lit.
- End on node 7 · Human with the logged decision chip.

Point at

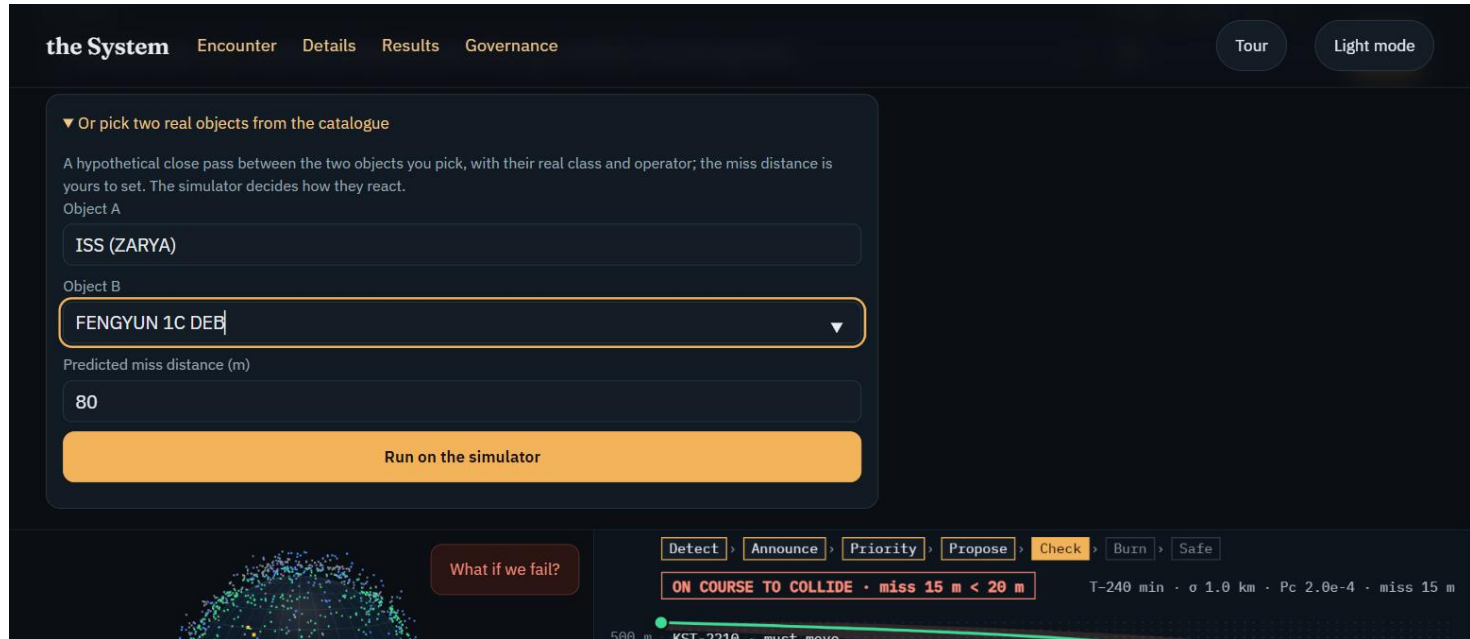
- 1 Sense: the risk bars. 2 Announce: the message chips.
- 3 Priority check: the lit pill (Crewed holds).
- 4 AI proposal: the probability bars. 5 Safety layer: ✓ allowed or ■ changed. 6 Execute: the burn. 7 Human: your approval, logged with the time.

Sav

“Every step is visible: what the AI senses, the messages, which priority check fired, what each AI proposed, what the safety layer allowed or changed, the burn, and the logged human decision.”

Beat 8 · 1:20 – 1:28

Encounter page · “Or pick two real objects” (team server address)



Do

- Open “Or pick two real objects from the catalogue”. Type ISS (ZARYA) and FENGYUN 1C DEB. Press Run on the simulator.
- A dialog opens because the station is crewed: approve it.

Point at

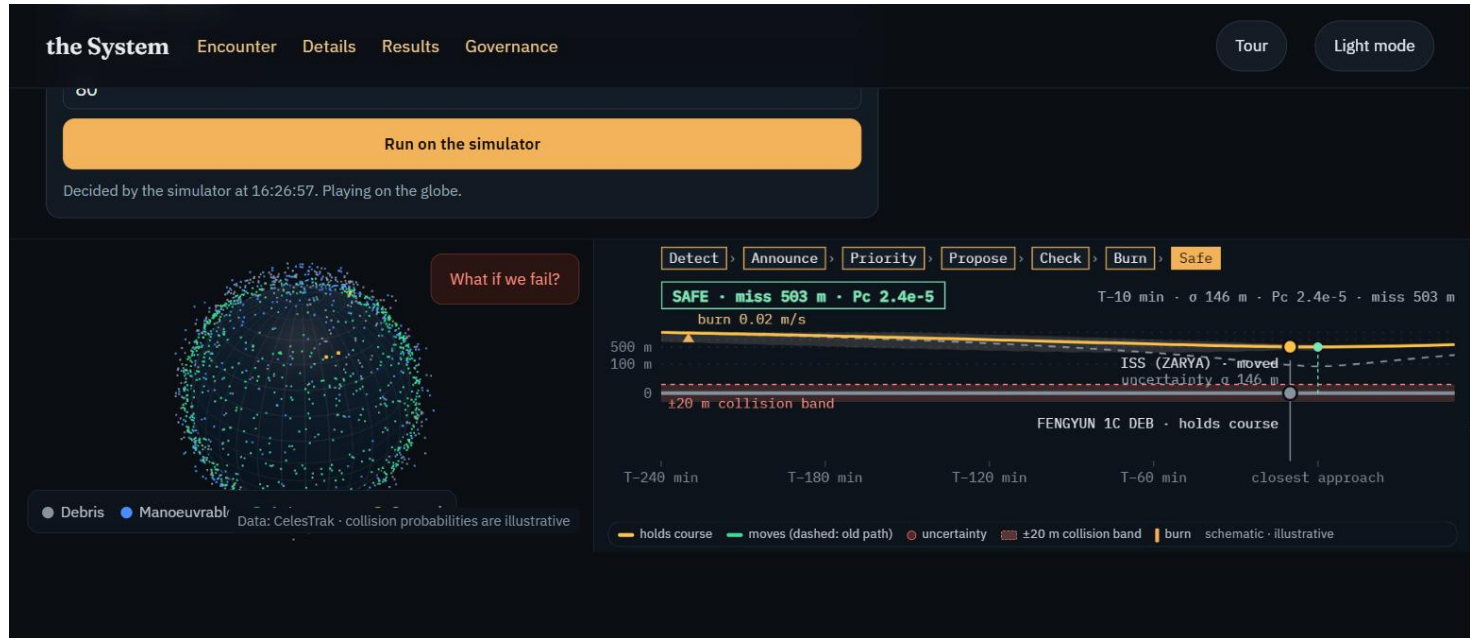
- The two names come from the real CelesTrak catalogue.
- The status line: Decided by the simulator at ...

Sav

“Pick any two real objects — the space station and a piece of debris — and the simulator decides how they would react.”

Beat 9 · 1:28 – 1:35

Encounter page · approach view for the real pair



Do

- Let it play to the end (or scrub the timeline to the last step).

Point at

- ISS (ZARYA) · moved; FENGYUN 1C DEB · holds course (debris cannot move).
- SAFE · miss 503 m at closest approach.
- Optional: on the Details page the Ledger tab shows the charge to the debris owner.

Sav

“Debris can't move, so the station moves, and the ledger charges the debris owner.”

Details page · Results (nav: Details › Results)

the System	Encounter	Details	Worked examples	Simulate	2019 replay	Results	How it works	Governance	Sources
Metric (95% CI)					Shared onboard AI (with safety layer)				
Accuracy — scenarios ending safely					99.34% (99.16–99.48)				
Recall — would-collide scenarios resolved					91.66% (89.52–93.39)				
Precision — burns in would-collide scenarios					7.91% (7.4–8.46)				
Collisions (avoidable / unavoidable)					50 / 16				
Unavoidable collisions are between two objects that cannot move (debris, silent or out of fuel). Precision is low by nature: under tracking uncertainty every risky pass needs a manoeuvre, and only some would really have collided.									
Strategy				Collisions %	Mean Δv (m/s)	Manoeuvres / event		Burden Gini	
Do nothing				7.9	0.000	0.00		0.00	

Do

- Open Details, click Results in the nav. Hold on the metrics table, then scroll to the strategy table and the collisions-per-strategy plot.

Point at

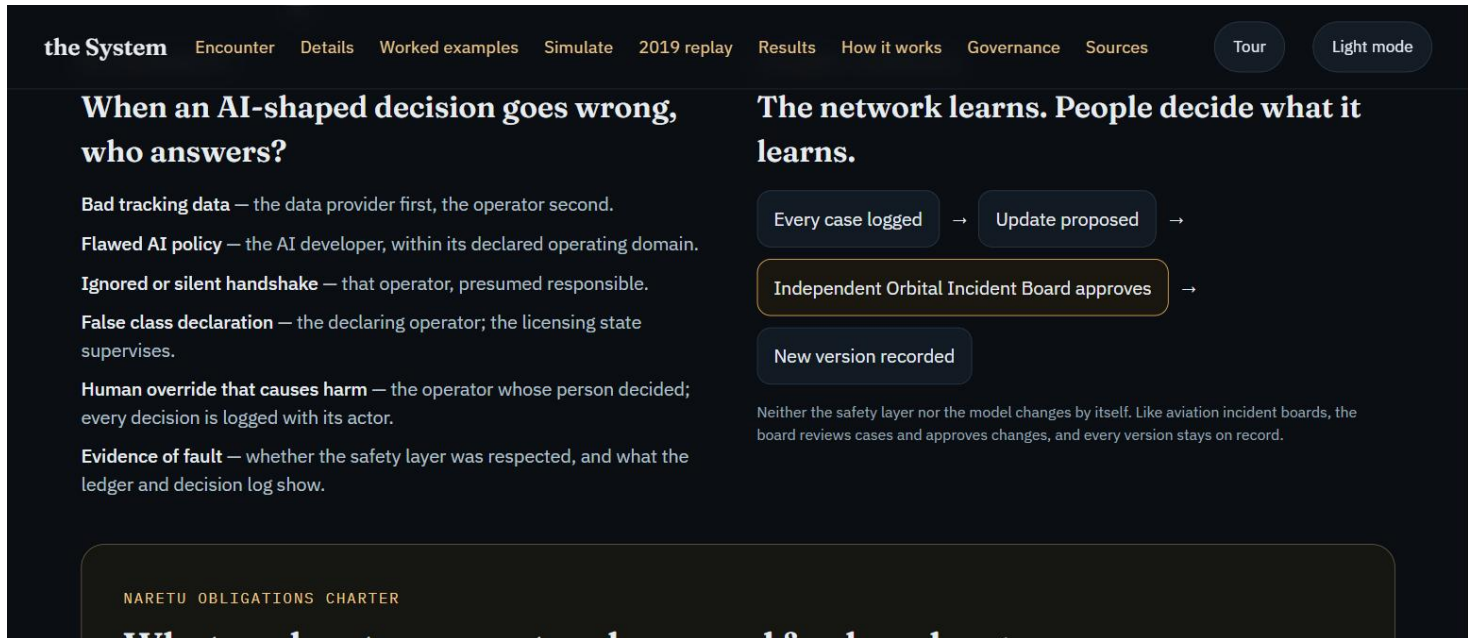
- Accuracy 99.34 % and Recall 91.66 % with their confidence intervals.
- The strategy table: 7.9 % collisions if nobody acts versus 0.7 % with the AI.

Sav

“We trained the AI on eight hundred million simulated encounters built from ESA's real conjunction data. On ten thousand new scenarios it ended ninety-nine point three percent safely, and resolved nine in ten that would otherwise have collided.”

Beat 11 · 1:48 – 1:58

Details page · Governance (nav: Governance)



Do

- Click Governance in the nav. Show the responsibility list and the learning chain, then scroll to the NARETU charter and end on “Everyone owes those not yet launched”.

Point at

- The six owners of failure.
- The chain: every case logged → update proposed → independent board approves → new version recorded.
- The last charter card.

Sav

“Humans stay accountable: every failure has an owner, and nothing in the network changes without an independent board. And following NARETU, everyone owes those not yet launched a usable orbit.”